SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37- SOFTWARE TESTING

EXPERIMENT 1

System Specification and Software Defect Identification – ATM System

AIM:

To study the system specifications of an ATM system, identify and analyze various software defects related to authentication, cash withdrawal, balance updates, transaction processing and session security, and prepare a defect report with severity and priority.

1. Problem Description

An ATM system allows customers to insert an ATM card, enter a PIN, check balance, withdraw cash, and receive transaction confirmation. The system must authenticate users, validate transaction amounts, update balances correctly, and handle hardware or network failures safely.

2. System Specifications

Spec ID	Function	Specification
S01	Card authentication	Valid and readable card is required before account access.
S02	PIN validation	Only the correct PIN should allow access.
S03	Balance inquiry	Displayed balance must match the account balance.
S04	Cash withdrawal	Amount must be positive and within balance and ATM limits.
S05	Cash dispensing	Cash must be dispensed only after successful authorization.
S06	Balance update	Balance must be updated correctly after successful withdrawal.
S07	Transaction record	Transaction history/receipt must contain correct details.
S08	Session security	Session must close after logout or timeout.

3. Software Defects Identified

Defect ID	Scenario	Expected Result	Actual Result	Severity	Priority
D01	Invalid PIN accepted	Access should be denied for wrong PIN	ATM permits access after invalid PIN	Critical	High
D02	Withdrawal exceeds balance	Transaction should be rejected	ATM accepts amount greater than balance	Critical	High
D03	Zero/negative withdrawal	Invalid amount should be rejected	ATM accepts invalid amount	High	High
D04	Balance reduced without cash	Transaction should be reversed/safely handled	Balance reduced although cash was not dispensed	Critical	High
D05	Incorrect balance display	Correct balance should be displayed	Old/incorrect balance displayed	High	High

D06	Session remains after timeout	Protected pages should require login	User can access account after timeout	High	High
D07	Receipt mismatch	Receipt should match transaction	Wrong amount/reference printed	Medium	Medium
D08	Duplicate debit after network retry	Only one transaction should be posted	Same transaction posted twice	Critical	High

4. Sample Defect Report

Field	**Details**
Bug ID	BR-ATM-001
Title	ATM permits withdrawal greater than available balance
Steps	Insert card → enter valid PIN → select withdrawal → enter amount greater than balance → submit.
Expected	Transaction rejected with insufficient balance message.
Actual	Transaction is accepted.
Impact	May cause incorrect account balance and financial loss.
Status	Open — awaiting developer investigation.

5. Conclusion

The ATM system specifications were examined and major functional and security-related software defects were identified. Proper validation, transaction control, session management, and error handling are required for reliable ATM operation.